CREDIT CARD FRAUD DETECTION VIA PENALIZED LOGISTIC REGRESSION AND CLASS REBALANCING DIAGNOSTICS

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ABSTRACT

Financial fraud detection presents significant statistical challenges due to extreme class imbalance, where legitimate transactions vastly outnumber fraudulent ones. In this project diagnostic models was built to detect fraudulent credit card transactions within high-dimensional, highly skewed transaction data (n= 284, 807). To overcome the limitations of standard accuracy metrics on imbalanced datasets, class rebalancing strategies are implemented and compared including Synthetic Minority Over-sampling Technique (SMOTE) and random under-sampling. Penalized Logistic Regression models were trained on rebalanced distributions and evaluated using Precision-Recall-AUC and ROC-AUC metrics. The optimized model achieved a classification accuracy of 94% with high recall, demonstrating effective fraud detection while controlling false-positive rates.

PROJECT OVERVIEW AND STATSTICAL CHALLENGE

Financial institutions process millions of transactions daily, with fraudulent activity representing a tiny fraction of total volume. Standard machine learning classifiers trained on unadjusted imbalanced datasets tend to yield trivial models that predict the majority class for all instances, resulting in high apparent accuracy but zero operational utility.
This project was structured around three core statistical objectives:

- Addressing extreme class skewness using statistical resampling techniques without introducing data leakage.
- Evaluating penalized classification models using precision-recall dynamics rather than unweighted accuracy metrics.
- Achieving a high fraud detection rate (Recall/ Sensitivity) to minimize financial loss while maintaining acceptable precision to prevent unnecessary transaction declines.

DATASET STRUCTURE AND DIAGNOSTICS

The dataset contained anonymized European cardholder transactions evaluated across 30 numerical features:

- Time: Seconds elapsed between each transaction and the first transaction in the dataset.
- V1 to V28: Principal components obtained via PCA due to confidentiality constraints.
- Amount: Transaction dollar value exhibiting strong right-skewness.
- Class: Binary target variable (0 = Legitimate, 1 = Fraudulent).

Key Diagnostic Findings:

- Extreme Class Imbalance: Out of 284,807 transactions, only 492 were fraudulent (0.172% minority class ratio), creating a 578:1 imbalance ratio.
- Feature Distributions: Several principal components (notably V10, V12, V14 and V17) exhibited significant distributional shifts between legitimate and fraudulent classes.
- Skewed Feature Scaling: Transaction amount varied drastically ($0.00 to $25,691.16) requiring robust standardization.

PREPROCESSING AND RESAMPLING PIPELINE

To handle scale variances and class imbalance without distorting statistical inference:

- Feature Scaling: Applied RobustScaler (utilizing median and interquartile range) to scale Amount and Time, mitigating the influence of extreme monetary outliers.
- Data Partitioning: Stratified the dataset into an 80% training set and a 20% test set prior to applying any resampling techniques, ensuring zero data leakage into the evaluation distribution.

The resampling strategies are applied strictly to training data

- Synthetic Minority Over-sampling Technique (SMOTE): Synthesized artificial minority instances along feature-space line segments connecting nearest neighbors.
- Random Under-Sampling: Down-sampled the majority class to balance class distributions for baseline comparison.
- Combined Sampling: Applied SMOTE to eliminate borderline noisy synthetic observations.

MODELLING METHODOLOGY AND EXPERIMENTAL RESULTS

Standard Classifier (Unadjusted Baseline):

- Accuracy: 99.83% (Misleading Metric)
- Fraud Recall: 58.2% (Missed over 40% of fraudulent cases)
- ROC-AUC: 0.884

Penalized Logistic Regression & SMOTE:

- Classification Accuracy:94.0%
- Fraud Recall (Sensitivity): 91.8%
- Precision-Recall AUC(PR-AUC):0.742
- ROC-AUC: 0.968

Key Analytical Observations:

- Accuracy Paradox: The unadjusted baseline model achieved near-perfect accuracy (99.83%) while missing over 40% of fraudulent transactions. Rebalancing sacrificed marginal overall accuracy to boost fraud recall to 91.8%.
- Metric Selection: Precision-Recall curves provided a far more informative diagnostic tool than standard ROC curves under heavy class imbalance, highlighting trade-offs in low-prevalence regimes.

- Feature Coefficients: Penalized logistic regression confirmed that features V14,V10 and V12 carried strong negative coefficients serving as major diagnostic indicators of fraudulent activity.

CONCLUSIONS:

- Class Rebalancing is essential: Statistical resampling (SMOTE) combined with cost-sensitive penalized regression is critical for detecting rare events in financial transaction streams.
- Leakage Prevention: Resampling must be strictly restricted to cross-validation training folds to ensure unbiased out-of-sample performance estimation.

Future Work: Future iterations will evaluate ensemble architectures(such as XGBoost and Isolation Forests) and incorporate cost- matrix optimization to directly balance the financial cost of missed fraud against the customer friction of false alarms.